

Patient Shortage in Histopathology: Correct Patient-Level Weights Do Not Rescue the Within-Patient Directions, so Their Direction Is the Bottleneck

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Abstract

The preceding study added the within-patient directions of a five-patient cohort to cohort whitening and gained 0.45 points on TCGA-UT, a sixth of the 2.66-point gap to whitening along the training pool’s basis, although the added directions hold 98 percent of the pool’s between-patient variance. It attributed the shortfall to the weights: the added directions carry patch-level variances, not patient-level ones. This report tests that explanation as a mechanism and upper-bound study, not as a remedy, by attaching to each added direction the training pool’s own between-patient variance along it. Two brackets separate weights from directions: random complement directions with the same oracle weighting, and the pool’s own best complement eigenvectors with their eigenvalues. On TCGA-UT, oracle weights leave the gain unchanged: the oracle-weighted arm exceeds cohort whitening by 0.44 points (95% interval: 0.27 to 0.61) and differs from the patch-weighted arm by -0.01 points (-0.11 to 0.10). Patch-level variance already ranks the directions almost exactly as the pool does (Spearman correlation 0.999), and the tuned total weight of the earlier study matches the oracle total. The pool’s best complement directions, in the same construction, recover 0.80 of the gap (0.71 to 0.88), 1.67 points more than the within-patient directions (1.37 to 1.99). BRACS gives the same ordering with wider intervals. Weights are therefore not the bottleneck; the within-patient directions are not the directions that matter for whitening, even though they lie in the right span.

1 Motivation

Whitening along the between-patient eigenbasis of a five-patient cohort gains 2.26 percentage points on TCGA-UT (Queisler, 2026a), whereas whitening along the training pool’s basis gains 4.92, a mechanism reference that reads all training patients. The 2.66-point difference is carried by directions outside the cohort basis rather than by the cohort’s eigenvalues (Queisler, 2026b). Queisler (2026c) asked whether the cohort’s own patch residuals reveal these directions. They do locate them: the top $8r_B$ eigenvectors of the expanded covariance hold 0.977 of the pool’s between-patient variance at five patients. The accuracy gain over cohort whitening was nevertheless 0.45 points, 17 percent of the gap, and direction-specific, since the same eigenvalues on random directions gained nothing.

That study explained the dissociation between span and accuracy by the weights. Whitening acts through the weight ν_j on each direction, and the added directions carried patch-level variances $\sigma\omega_j$ where the pool basis carries patient-level variances. Its limitations section named the test that was not run: the same directions with the pool’s between-patient variance attached to each. This report runs it.

The question is: *once each within-patient direction carries the pool’s between-patient variance, does the gain approach that of pool-basis whitening?* The answer is a mechanism, not a remedy,

because the weights read the training pool and no cohort can compute them. The study is judged against the readings fixed in Section 4, at five patients per class, separately on each dataset.

2 Estimator model

The setting and notation follow Queisler (2026c). The cohort’s between-patient covariance is $\hat{\Sigma}_B = \mathbf{U}^\top \text{diag}(\boldsymbol{\lambda})\mathbf{U}$ with $r_B \leq C(G - 1)$ eigenvectors, and the within-patient complement $\mathbf{V}$ collects the eigenvectors of the within-patient patch covariance projected off $\mathbf{U}$, ordered by the patch-level variance $\boldsymbol{\omega}$. Whitening transforms every feature vector by

$$\mathbf{x} \mapsto \mathbf{x}(\mathbf{I} + \kappa \mathbf{B}^\top \text{diag}(\boldsymbol{\nu}) \mathbf{B})^{-1/2}, \quad (1)$$

with basis $\mathbf{B}$ and spectrum $\boldsymbol{\nu}$ (Queisler, 2026a). Let Σ_{pool} be the between-patient covariance of the training pool. The *oracle weight* of a unit direction $\mathbf{v}$ is the pool’s between-patient variance along it, $\mathbf{v}^\top \Sigma_{\text{pool}} \mathbf{v}$. All arms of this study use the basis $\mathbf{B} = [\mathbf{U}; \mathbf{A}]$ with the cohort spectrum $\boldsymbol{\lambda}$ on the cohort block and oracle weights on the added directions $\mathbf{A}$; there is no trace scale s , because the oracle weights fix the scale. Three choices of $\mathbf{A}$ are compared:

- **Wo:** $\mathbf{A} = \mathbf{V}$, the cohort’s within-patient complement, with oracle weights $\mathbf{v}_j^\top \Sigma_{\text{pool}} \mathbf{v}_j$ in place of $s\tau\omega_j$.
- **Ro:** $\mathbf{A}$ is the same number of random orthonormal directions orthogonal to $\mathbf{U}$ (the draw of the preceding study), with oracle weights. Wo–Ro is the value of the within-patient directions once the weights are correct.
- **Po:** $\mathbf{A}$ collects the leading eigenvectors of $\mathbf{P}_\perp \Sigma_{\text{pool}} \mathbf{P}_\perp$, with $\mathbf{P}_\perp = \mathbf{I} - \mathbf{U}^\top \mathbf{U}$, and their own eigenvalues as weights. These are the best directions and weights that the construction “cohort block plus complement” can use. Po–Wo is the direction loss of $\mathbf{V}$, and RW–Po is the cost of keeping the cohort block and its spectrum in place of the pool’s.

The oracle weight is a diagonal entry. The pool covariance also has off-diagonal terms between the directions of $\mathbf{V}$, which the whitening basis ignores unless $\mathbf{V}$ diagonalizes $\mathbf{P}_\perp \Sigma_{\text{pool}} \mathbf{P}_\perp$. Wo therefore tests weights and directions jointly, and Ro and Po bracket it. The scale factor κ is anchored to the cohort block as before: grid factors are multiplied by the ratio of the mean eigenvalue of the expanded spectrum to the mean cohort eigenvalue, so that each factor applies the same κ to the cohort block as in cohort whitening (Queisler, 2026c).

3 Design

Data, features, splits, cohort draws, and classifier follow Queisler (2026c): TCGA-UT with 30 cancer types and BRACS with seven lesion classes, frozen 2,560-dimensional Virchow2 features (Zimmermann et al., 2024), three patient-disjoint splits, ten draws per split, and multinomial logistic regression with validation-selected regularization. Only the five-patient cohorts with 32 patches per patient are fitted, so each dataset needs 30 fits per arm. The random directions of Ro use the generator and seed of the preceding study, so they are the same directions as in its control.

Table 1 lists the arms. Every fitted arm tunes $\kappa \in \{0.1, 1, 10, 100\}$, rescaled as above, jointly with the regularization strength on validation patient-macro balanced accuracy. The reused arms are the exact fits of the source studies on the same cohorts, so all contrasts are paired draw by draw.

Arm	Added directions	Added weights	Role
RWc5	none	—	Cohort whitening (reused)
RW5	pool basis	pool	Pool-basis whitening (reused mechanism reference)
Wt5	within-patient $\mathbf{V}$	$s\tau\omega_j$, tuned s	Patch-level weights (reused from Experiment 23)
Rt5	random, $\perp \mathbf{U}$	$s\tau\omega_j$, tuned s	Random control of the preceding study (reused)
Wo5	within-patient $\mathbf{V}$	pool variance along $\mathbf{v}_j$	Oracle weights on the cohort’s directions
Ro5	random, $\perp \mathbf{U}$	pool variance along $\mathbf{a}_j$	Oracle weights on random directions
Po5	top eigenvectors of $\mathbf{P}_\perp \boldsymbol{\Sigma}_{\text{pool}} \mathbf{P}_\perp$	their eigenvalues	Best directions and weights in the complement

Table 1: All new arms keep the cohort block and its spectrum. They differ in which directions are added and how these are weighted; all weights read the training pool, so the new arms are mechanism references and not remedies.

4 Evaluation

The endpoint is patient-macro balanced accuracy on the test partition, pooled over splits and draws with equal weight per split. Uncertainty combines Bayesian bootstrap weights on test patients (Rubin, 1981) with resampling of draws over 2,000 replicates shared by all arms, so every contrast is paired; intervals are 95 percent percentile intervals.

The primary contrast is Wo5–Wt5 with a one-point threshold: oracle weights count as the missing ingredient if they add at least one point over patch-level weights with an interval above zero. The secondary contrasts are Wo5–Ro5, Po5–Wo5, and RW5–Po5, and the headroom share $(x - \text{RWc5})/(\text{RW5} - \text{RWc5})$ of each arm x . Four readings were fixed in advance.

1. Wo $\gg$ Wt and Wo $\approx$ Po: the cohort’s directions are usable and missing patient-level weights are the bottleneck.
2. Wo $\approx$ Wt and Po $\gg$ Wo: the within-patient directions are wrong at the resolution whitening needs, even when correctly weighted.
3. Po $\ll$ RW: the construction itself caps the gain; headroom needs the pool’s full basis.
4. Ro $\approx$ Wo: the gain comes from oracle weights alone and directions are irrelevant.

Diagnostics are the selected κ factors, the added rank, the oracle-implied scale $\sum_j \nu_j / \sum_j \lambda_j$ over the added directions (comparable with the selected s of the preceding study), the added share of the pool’s between-patient variance, and, for Wo, the Spearman correlation between the patch-level variance ω_j and the oracle weight along $\mathbf{V}$.

5 Results

The results follow the argument: the accuracy ladder from cohort whitening to pool-basis whitening, the contrasts that separate weights from directions, and the diagnostics that explain them.

5.1 Accuracy ladder

Oracle weights do not lift the within-patient directions on either dataset (Table 2). On TCGA-UT the oracle-weighted arm Wo5 reaches 65.18 percent, 0.44 points above cohort whitening (0.27 to 0.61) and 0.01 points below the patch-weighted arm Wt5 of the preceding study (−0.11 to 0.10). The primary contrast is therefore zero to within 0.11 points, far from the one-point threshold. Wo5 closes a share of 0.166 (0.107 to 0.228) of the 2.66-point gap between cohort-basis and pool-basis whitening, the same as the 0.169 of Wt5. On BRACS the point estimates are the same order: Wo5 gains 0.32 points over cohort whitening (−0.31 to 0.94) and lies 0.13 points below Wt5 (−0.73 to 0.42), with a share of 0.106 (−0.126 to 0.324) of the 3.01-point gap.

The pool’s own complement directions behave entirely differently. Po5 reaches 66.86 percent on TCGA-UT, 2.12 points above cohort whitening (1.78 to 2.46) and 1.67 points above Wo5 (1.37 to 1.99), and closes a share of 0.796 (0.714 to 0.880) of the gap. On BRACS it gains 2.33 points over cohort whitening (0.97 to 3.88), 2.01 over Wo5 (0.74 to 3.38), with a share of 0.773 (0.482 to 1.031). Po5 stays below the full pool-basis whitening RW5 by 0.54 points on TCGA-UT (0.31 to 0.77) and by 0.68 on BRACS (−0.08 to 1.62), so the sign is the expected one and the cost of keeping the cohort block and spectrum is small.

The random directions with oracle weights, Ro5, gain 0.02 points on TCGA-UT (−0.00 to 0.05) and 0.05 on BRACS (−0.02 to 0.21), the same as the random control of the preceding study. Oracle weighting alone therefore does nothing, and the 0.42-point advantage of the within-patient directions over random ones persists: Wo5−Ro5 is 0.42 points on TCGA-UT (0.26 to 0.59) and 0.27 on BRACS (−0.35 to 0.88).

	TCGA-UT	BRACS
Cohort whitening (RWc5)	64.74 [63.52, 65.90]	36.49 [33.87, 39.24]
Patch-level weights (Wt5)	65.19 [63.95, 66.34]	36.94 [34.39, 39.48]
Oracle weights, within-patient (Wo5)	65.18 [63.94, 66.32]	36.81 [34.18, 39.46]
Oracle weights, random (Ro5)	64.76 [63.53, 65.93]	36.54 [33.89, 39.26]
Best complement directions (Po5)	66.86 [65.62, 67.99]	38.82 [36.10, 41.27]
Pool-basis whitening (RW5)	67.40 [66.14, 68.52]	39.50 [36.82, 41.98]
Wo5 − RWc5	0.44 [0.27, 0.61]	0.32 [−0.31, 0.94]
Wo5 − Wt5	−0.01 [−0.11, 0.10]	−0.13 [−0.73, 0.42]
Wo5 − Ro5	0.42 [0.26, 0.59]	0.27 [−0.35, 0.88]
Po5 − Wo5	1.67 [1.37, 1.99]	2.01 [0.74, 3.38]
Po5 − RWc5	2.12 [1.78, 2.46]	2.33 [0.97, 3.88]
RW5 − Po5	0.54 [0.31, 0.77]	0.68 [−0.08, 1.62]
Headroom share, Wt5	0.169 [0.105, 0.231]	0.149 [−0.140, 0.391]
Headroom share, Wo5	0.166 [0.107, 0.228]	0.106 [−0.126, 0.324]
Headroom share, Ro5	0.007 [−0.001, 0.018]	0.015 [−0.009, 0.077]
Headroom share, Po5	0.796 [0.714, 0.880]	0.773 [0.482, 1.031]

Table 2: Correct weights leave the within-patient directions where they were, while the pool’s own complement directions recover about four fifths of the headroom. Accuracy entries are test patient-macro balanced accuracy in percent at five patients per class, contrasts are paired differences in percentage points, and the headroom share is $(x - \text{RWc5}) / (\text{RW5} - \text{RWc5})$. Intervals are 95% bootstrap intervals over test patients and draws. The bold row is the primary contrast. RWc5, RW5, Wt5, and Rt5 are reused from the source studies.

5.2 Diagnostics

The patch-level variance already ranks the added directions as the pool does (Table 3). Across the directions of $\mathbf{V}$ on TCGA-UT, the Spearman correlation between ω_j and the oracle weight is 0.999 on average over the 30 fits, and 0.981 on BRACS. The total weight matches as well. The oracle-implied scale of Wo5 is 0.296 on TCGA-UT, equal to the preceding study’s selected values $s = 0.3$ (12 of 30 fits) and $s = 1$ (13 of 30), and the added directions carry 30.2 percent of the pool’s between-patient variance in Wo5 and Po5 alike, because on TCGA-UT both bases fill the complement of the cohort basis (rank 2,438 to 2,440). The oracle weights of Wo5 and Po5 thus have the same total, spread over different directions. On BRACS the implied scale is 0.412 for Wo5, near the selected values 0.3 to 3 of the preceding study.

The random arm Ro5 has the same rank and total on TCGA-UT (0.296, 30.1 percent), but not the same weights within the complement, and it gains nothing. On BRACS, where $\mathbf{V}$ has rank 1,085 and does not fill the complement, the random directions carry only 18.1 percent of the pool variance against 40.1 for Wo5 and 42.3 for Po5, which needs 227 directions. Po5 reaches a comparable share with a fifth of the directions.

The selected whitening scale reflects how much variance the arm adds. On TCGA-UT Po5 selects factor 100 in 29 of 30 fits, Wo5 factor 10 in 21 (1 in 5, 100 in 4), and Ro5 factor 1 in 23 (10 in 7). Since the factors are anchored to the cohort block, the classifier contracts more strongly the more of the added block lies along directions that matter. On BRACS the factors spread: Po5 selects 10 or 100 in 24 of 30 fits, Wo5 spreads over 0.1 to 10 (12, 7, 10) with one fit at 100, and Ro5 selects 0.1 in 18.

	TCGA-UT			BRACS		
	Wo5	Ro5	Po5	Wo5	Ro5	Po5
Added rank	2,438	2,438	2,440	1,085	1,085	227
Added share of pool variance	0.302	0.301	0.302	0.401	0.181	0.423
Oracle-implied scale	0.296	0.296	0.296	0.412	0.186	0.435
Spearman(ω_j , oracle weight)	0.999	—	—	0.981	—	—
Selected κ factor 0.1	0	0	0	12	18	4
Selected κ factor 1	5	23	0	7	6	2
Selected κ factor 10	21	7	1	10	4	10
Selected κ factor 100	4	0	29	1	2	14

Table 3: The patch-level variance of the within-patient directions matches the oracle weights in rank and total, so reweighting cannot be what limits Wt5. Entries are means over 30 fits at five patients; the oracle-implied scale is the added weight relative to the cohort’s between-patient trace, comparable to the selected s of the preceding study; the κ rows count fits.

6 Discussion

The prespecified readings select the second, with the third in mild form. Wo5 equals Wt5 to within 0.11 points on TCGA-UT, while Po5 exceeds Wo5 by 1.67 points, so the patient-level weights are not what limited the within-patient directions. The diagnostics agree: patch-level variance already ranks the directions almost exactly as the pool does, and its tuned total equals the oracle total. The first reading, that missing weights are the bottleneck, is rejected, and the explanation given by Queisler (2026c) does not hold. The fourth reading is rejected as well: Ro5 gains nothing, and Wo5–Ro5 is 0.42 points on TCGA-UT, so the gain does not come from oracle weights alone.

What limits the within-patient directions is their orientation. The complement they fill is the same as that of Po5 on TCGA-UT, and both add the same total weight, but Po5 aligns the added directions with the eigenvectors of the pool’s between-patient covariance, and Wo5 does not. The span measure of the preceding study, the share of pool variance in the span of the added directions, cannot detect this. On TCGA-UT it is saturated for any basis that fills the complement, and whitening acts on the eigenstructure within the span. A direction $\mathbf{v}_j$ can carry a large diagonal weight $\mathbf{v}_j^\top \Sigma_{\text{pool}} \mathbf{v}_j$ and still mix several pool eigenvectors; the whitening then contracts a mixture of directions that are not separately worth contracting by that amount. The construction of Po5, which places the eigenvectors themselves, avoids this. This is an interpretation of the contrast between Wo5 and Po5, not a separate measurement, but it is the only difference between the two arms.

Two consequences follow for cohort-only remedies. The within-patient covariance identifies the right subspace and a suitable scale, but its eigenvectors do not diagonalize the patient-level covariance, and no reweighting repairs that. And the ceiling of the construction is high: Po5 recovers 0.80 of the gap on TCGA-UT while keeping the cohort’s own block and spectrum, so most of the headroom lies in the orientation of the complement, not in the cohort block. Any cohort-only estimate of that orientation would need patient-level information about the complement that the cohort’s $C(G-1)$ patient deviations do not contain. Unlabelled slides from further patients, related cohorts, or a population covariance carried by the feature extractor are candidate sources, as before.

7 Limitations

All new arms read the training pool, so the results are a mechanism and upper bound, not evidence of a usable estimator. Po5 in particular uses the pool’s eigenvectors and cannot be computed from a cohort.

The oracle weight is the diagonal entry $\mathbf{v}_j^\top \Sigma_{\text{pool}} \mathbf{v}_j$. It ignores off-diagonal pool covariance between the directions of $\mathbf{V}$, and that is exactly what the contrast with Po5 measures; an arm that rotates $\mathbf{V}$ toward the pool eigenbasis in stages would show how much orientation is needed, but was not run. The study fixes the cohort block and its spectrum and keeps the cohort spectrum on the block, so Po5 and RW5 differ by the block only; the 0.54-point difference on TCGA-UT is therefore the cost of that choice and not of the added directions.

Only five patients per class were fitted, since the gain of the preceding study was gone at ten and twenty patients. On BRACS the complement has about a thousand directions instead of filling the space, so Wo5 and Po5 differ in span as well as in orientation there, and the intervals of the BRACS contrasts are wide: Wo5–Wt5 cannot exclude effects of about half a point in either direction, and the Po5 headroom share has an interval reaching above one. The hypotheses were fixed before the fits, but all arms reuse the three evaluation splits of the source studies. The results are conditional on frozen Virchow2 features, multinomial logistic regression, 160 patches per class, and the two cohorts studied.

References

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